Multi-Signal Fusion for Robust AI-Generated Text Detection

Morphy (Fei) Mo **Hanyu Gong** **Wayne Zhang** **Yazhen Li**
Department of Computer Science, College of Arts and Science
New York University
{fm2503, hg2811, lz3080, yl11087}@nyu.edu

Abstract

This paper studies robust AI-generated text detection under generator and domain shift. We build a multi-signal detection framework that combines three complementary sources of evidence: lexical TF-IDF features, statistical signal features, and contextual representations from a fine-tuned RoBERTa classifier. To combine these detectors, we evaluate both single-layer fusion methods and a hierarchical ensemble.

By evaluating our system on a self-constructed Controlled dataset, HC3, and a balanced English subset of M4, we show three key findings. First, contextual transformer representations provide the strongest standalone detection performance, but they still degrade under unseen generator shift. Second, statistical signal features are weaker as standalone detectors, but they provide useful complementary evidence for fusion. Third, fusion improves robustness under distribution shift by combining lexical, statistical, and contextual signals. Overall, our results suggest that robust AI-text detection should rely on multiple complementary representations rather than a single detector.

1 Introduction

With the advancement of LLM, distinguishing between human-written and AI-generated text has become increasingly challenging. AI detection becomes essential in protecting Academic Integrity, improving Content Moderation, avoiding misinformation usage, and preserving AI Transparency.

The recent paper of human-written and AI-generated text detection focuses on structuring models together to improve and compare the detection result. In this paper, we propose an adaptive hybrid AI-text detection system that combines lexical, statistical, and fine-tuned transformer signals. We have built two other baseline models. Logistic regression model with TF-IDF and logistic regression model with signal features. The two baseline models learned statistical and lexical parts of the models. We created a fine-tuned RoBERTa model that captures deep semantic understandings of text. All three models have a different perspective of understanding. Therefore, we decided to ensemble these models to improve the robustness of the model.

The ensemble method we use is single-layer and hierarchical ensemble methods. The system estimates signal reliability and adjusts the contribution of each module. This design addresses the limitation that statistical signals work well in controlled settings but degrade under cross-dataset and cross-generator shifts, and fine-tuned RoBERTa model works well in controlled settings but degrades under cross-generator shifts.

2 Related Work

Prior work on AI-generated text detection has explored statistical, likelihood-based, and neural approaches. GLTR (Gehrmann et al., 2019) visualizes token-level language model likelihoods to expose generation artifacts, while DetectGPT (Mitchell et al., 2023) detects machine-generated text using probability curvature. These methods motivate our use of likelihood-based signal features such as perplexity, average log probability, and token-level burstiness.

Benchmark datasets have also shaped this task. HC3 (Guo et al., 2023) compares human-written and ChatGPT-generated answers, while M4 (Wang et al., 2024) provides a larger multi-generator and multi-domain benchmark. Prior work has shown that detectors often perform well in-domain but degrade under unseen domains, unseen generators, or evasion tactics (Chakraborty et al., 2023; Pudasaini et al., 2025). This motivates our evaluation design, which includes in-domain, held-out generator, and held-out domain settings.

Recent work has also explored combining stylis-

tic and neural information. MoSEs (Wu et al., 2025) uses a mixture of stylistic experts for uncertainty-aware AI-text detection. Our work follows this general direction but focuses on a simpler and more transparent fusion framework: we combine lexical TF-IDF, statistical signal features, and fine-tuned RoBERTa predictions using both single-layer and hierarchical fusion methods.

3 Methodology

3.1 Baselines

To provide transparent and reproducible reference points for human-vs-AI text classification, we implement two complementary classical baselines that represent distinct inductive biases. The first is a lexical baseline A utilizing sparse n-gram features, which prioritizes semantic content (i.e., what is being said). Complementing this, the second is a signal-based baseline B driven by lightweight statistical descriptors (e.g., perplexity, burstiness, etc.), which captures stylistic signatures (i.e., how it is being expressed) through structural patterns. Together, these baselines offer a multi-dimensional framework for interpreting the discriminative boundaries between human and AI text.

3.1.1 Baseline A: TF–IDF + Logistic Regression

This TF–IDF-based baseline A follows the standard strong linear-text-classification recipe, with the inductive bias that certain words, collocations, and short phrases occur with systematically different frequencies in human vs. AI text.

We construct a scikit-learn pipeline that (i) extracts the raw text field from each JSON example, (ii) computes TF–IDF features over word unigrams and bigrams, and (iii) trains a logistic regression classifier. For reducing noise from rare n-grams, we cap the vocabulary at 20,000 features. For ensuring meaningful contribution, we also remove English stopwords (e.g., very frequent function words "*the*", "*a/an*", "*to*", etc.)

3.1.2 Baseline B: Signal Features + Logistic Regression

The signal baseline represents each text with handcrafted statistical and stylometric features rather than lexical tokens. We use seven features: perplexity, average log probability, LM-based burstiness, sentence-length burstiness, type-token ratio, punctuation ratio, and text length. These features capture predictability, local variation, lexical diversity, formatting style, and length-related effects.

We standardize all features using `StandardScaler` before training the same class-balanced logistic regression classifier. This prevents large-scale features such as perplexity from dominating smaller-scale features such as punctuation ratio.

Because these features have different numerical scales, we standardize them using `StandardScaler` before training the same class-balanced logistic regression classifier. This prevents high-magnitude features, such as perplexity, from dominating the optimization. We apply StandardScaler to standardize each feature dimension to zero mean and unit variance, then train the same class-balanced logistic regression classifier. Standardization is essential here because the signals operate on very different numeric scales (e.g., perplexity can be tens-to-hundreds, while punctuation ratio is typically a small number in ([0,1])). Without scaling, large-magnitude features can dominate optimization and interact unfairly with L2 regularization, degrading performance.

3.1.3 Baseline C: Fine-tuned RoBERTa Classifier

This baseline represents the contextual modeling perspective in our system. We fine-tune `roberta-base` (125M parameters) for binary human-versus-AI classification using HuggingFace Transformers and PyTorch. Each input text is tokenized with the matching `roberta-base` tokenizer, truncated to a maximum length of 512 tokens, and passed through the transformer encoder. A binary classification head is trained on top of the final representation to predict whether the text is human-written or AI-generated.

We fine-tune RoBERTa mainly on the M4 dataset using a 70/15/15 train/dev/test split. For out-of-distribution evaluation, we also construct held-out generator and held-out domain settings: in the held-out generator setting, ChatGPT samples are excluded from training and used only for testing; in the held-out domain setting, Wikipedia samples are excluded from training and used only for testing. These splits allow us to evaluate whether the fine-tuned neural detector generalizes beyond the generators and domains seen during training.

For training, we use AdamW with a learning rate of 2×10^{-5}, batch size 16, weight decay 0.01, warmup ratio 0.1, and train for 3 epochs with ran-

dom seed 42. Since the M4 subset is balanced between human and AI examples, we do not apply additional class weights. The best checkpoint is selected based on development F1. For the M4-to-M4 setting, the best checkpoint is `checkpoint-1258` under `results/roberta_results_M4_to_M4/`.

During inference, RoBERTa outputs logits that are converted into probabilities using softmax. We use the resulting AI-generation probability, $p(\mathrm{AI})$, both as the RoBERTa prediction and as an input feature for the fusion models.

3.2 Ensemble and Fusion Methods

In order to improve the quality of outputs of our task beyond the baseline models, we decided to ensemble based on the baseline models we mentioned in the first section and the RoBERTa model in addition. In this case, we included both single layer ensemble and double layer ensemble(also named as multi-hierarchical ensemble in our case). We include 3 types of single layer ensembles - Naive Average, Simple Stacking, and Signal-Aware MLP Gate - and one double layer ensemble. These 4 ensembles help achieve better output results comparing to the single baseline models in the previous section for our task of detecting AI and human written texts.

The three base detectors capture different views of the text: TF-IDF captures lexical patterns, signal features capture statistical and stylometric regularities, and RoBERTa captures contextual representations. Since these models have complementary strengths and failure modes, we evaluate ensemble methods that combine their predictions to improve robustness under distribution shift.

We consider two categories of ensemble methods. The first category consists of single-layer fusion methods, including naive averaging, simple stacking, and a signal-aware MLP gate. The second category is a double-layer hierarchical ensemble, which introduces an intermediate voting layer before the final prediction.

3.2.1 Single Layer Ensembles: Naive Average, Simple Stacking, and Signal-Aware MLP Gate (MLP Gate)

For single layer ensemble, we mainly ensemble three models TF-IDF, Signal Feature, and RoBERTa with three different methods: Naive Average, Simple Stacking, and Signal-Aware MLP Gate.

We evaluate three single-layer ensemble methods: Naive Average, Simple Stacking, and Signal-Aware MLP Gate. All three methods use the predicted AI probabilities from the TF-IDF, Signal, and RoBERTa models as base inputs.

Naive Average The naive average ensemble computes the final AI probability by averaging the three base-model probabilities:

$$p_{\text{final}} = \frac{p_{\text{tfidf}} + p_{\text{signal}} + p_{\text{roberta}}}{3}.$$

This method does not introduce any additional trainable parameters and serves as a simple fusion baseline.

Simple Stacking Simple stacking trains a logistic regression meta-classifier on the development-set predictions of the three base models. The input to the meta-classifier is the probability vector

$$[p_{\text{tfidf}}, p_{\text{signal}}, p_{\text{roberta}}].$$

At test time, the trained meta-classifier predicts the final label using the corresponding test-set probability vector.

Signal-Aware MLP Gate (MLP Gate) The Signal-Aware MLP Gate extends simple stacking by adding the signal features as auxiliary inputs. In addition to the three base-model probabilities, the MLP receives statistical and stylometric features such as perplexity, average log probability, LM-based burstiness, sentence-length burstiness, type-token ratio, punctuation ratio, and text length. This allows the model to learn non-linear interactions between neural predictions and signal-based reliability cues. The goal is to adaptively combine lexical, statistical, and contextual evidence rather than assigning fixed weights to each detector.

3.2.2 Double Layer Ensemble

The Double Layer Ensemble takes 9 individual models and ensemble then with 2 layers to get the output. The 9 individual models includes 3 pairs of TF-IDF and Signal Feature models and 3 RoBERTa models.

Each pair of TF-IDF and Signal Feature models are ensemble into a binary voter with different seeds - 42, 43, 44 - in specific for diversification - and this is of the first layer. Each binary voter is built with a Sequential neural network of 2 layers - one layer with linear transformation that takes 2 inputs and output 8 neurons, one layer with ReLU

activation, drop 30% of the ReLU outputs, and then has one layer that linearly transform 8 neurons and output 1 neuron, and eventually has one sigmoid activation for eventual output. In addition, 3 RoBERTa models, each with a distinct seed of 42, 43, 44 are included and ensemble along with the three binary ensembles - a total of 6 modules - into a meta voter for the second layer ensemble.

The meta voter is a Sequential Neural Network of 5 layers: one layer of linear transformation of the number of input dimension(6 here) and 8 output neuron. activated by ReLU and zeroed out 20% of it, another lnear layer of 8 inputs and 4 output neurons, also activated by ReLU and dropped 20% out, and the last layer that take the 4 inputs and output 1 neuron, eventually activated by sigmoid.

With the 9 to 6 to 1 construction of double layer ensemble, the whole structure is able to take in diverse models and results from them to produce a comprehensively high quality output.

4 Experiments

4.1 Datasets and Domains

We use three datasets in this project: a self-constructed Controlled dataset, HC3 (Guo et al., 2023), and the English subset of M4 (Wang et al., 2024). These datasets differ in domain coverage, generator diversity, and the degree of experimental control, allowing us to evaluate AI-generated text detection under both clean and heterogeneous settings.

The Controlled dataset is constructed by our group to reduce common dataset artifacts such as length imbalance and topic mismatch. It contains approximately 800 texts across essay- and article-style domains. For each AI-generated text, we pair a human-written text on a similar topic, and all samples are constrained to approximately 400–700 words. The AI texts are generated using GPT-4o-mini (OpenAI, 2024) under a natural undergraduate student writing style, while the human texts are collected from Wikipedia-style writing, news articles, and student essay-style sources. This dataset provides a cleaner setting for examining stylistic and statistical differences between human-written and AI-generated text.

HC3 (Guo et al., 2023) is a human–ChatGPT comparison corpus consisting of human-written and ChatGPT-generated answers to similar questions. We use approximately 13,000 English samples from HC3. Since HC3 focuses primarily on question-answering data generated by a single model, it serves as an intermediate benchmark between our highly controlled dataset and the more diverse M4 dataset.

Dataset	Size	Domains	Generator(s)
Controlled	∼800	Essay / article-style	GPT-4o-mini
HC3	∼13K	QA	ChatGPT
M4	14,370	5 domains	5 generators

Table 1: Summary of the datasets used in this project.

M4 (Wang et al., 2024) is the main large-scale benchmark used in our project. We use a balanced English subset of 14,370 texts with a 1:1 ratio of human-written and AI-generated samples. The subset covers five domains: Wikipedia, WikiHow, Reddit ELI5, ArXiv, and PeerRead. It also includes outputs from five generators: text-davinci-003, ChatGPT, Cohere, Dolly-v2, and BLOOMZ. Compared with the Controlled dataset and HC3, M4 provides substantially greater diversity in both writing domains and generation sources.

4.2 Evaluation

A central challenge in AI-generated text detection is distribution shift: detectors trained on one generator or domain may learn patterns that do not transfer to new generators or writing domains. To evaluate both classification performance and robustness, we design the experiment work by testing lexical baseline (TF-IDF), stylometric/statistical baseline (Signal features), neural baseline (fine-tuned RoBERTa), and fusion models under three settings: in-domain evaluation, held-out generator evaluation, and held-out domain evaluation.

4.2.1 In-domain evaluation.

We first measure standard in-domain performance by training, developing, and testing on the same dataset distribution. For Controlled and HC3, we construct stratified 70/15/15 train/dev/test splits with a fixed random seed of 42 to preserve label proportions. For M4, we use the same train/dev/test split consistently across all models. In-domain evaluation establishes an upper-bound setting for each dataset and allows us to compare how lexical, statistical, and contextual features behave when the train and test distributions are matched.

4.2.2 Held-Out Generator Evaluation

To test cross-generator robustness, we perform leave-one-generator-out evaluation on M4. The

held-out generators include `chatgpt`, `davinci003`, `cohere`, `dolly`, and `bloomz`. For each generator g, we train on examples from all other generators and test only on examples generated by g. This setting measures whether a detector can generalize to an unseen generator rather than memorizing generator-specific artifacts.

4.2.3 Held-Out Domain Evaluation

To test cross-domain robustness, we perform leave-one-domain-out evaluation on M4. The held-out domains include `wikipedia`, `reddit_eli5`, `wikihow`, `arxiv`, and `peerread`. For each held-out domain d, we train on all domains except d and test only on examples from d. This setting measures whether a detector relies on domain-specific vocabulary or captures signals that transfer across writing styles and topics.

4.2.4 Feature consistency.

For signal-based models, we standardize feature definitions across all datasets and splits to ensure fair comparison. In particular, perplexity, average log probability, LM-based burstiness, sentence-length burstiness, type-token ratio, punctuation ratio, and text length are computed with consistent definitions using `distilgpt2` as the unified reference language model and stored using a unified feature schema. This ensures that performance differences reflect model behavior rather than incompatible feature representations.

Together, these settings allow us to evaluate not only whether a detector performs well in-domain, but also whether it remains reliable when the generator or domain changes.

4.3 Metrics

We report accuracy, precision, recall, F1, Macro F_1, and ROC-AUC when probability scores are available. Since our task is binary classification between human-written and AI-generated text, the standard F1 score is computed for the AI-generated class. We use Macro F_1 as the primary metric because it averages performance across both classes and better reflects balanced classification quality, especially under held-out generator and held-out domain settings.

Together, these settings allow us to evaluate not only whether a detector performs well in-domain, but also whether it remains reliable when the generator or domain changes.

Dataset	Model	Acc.	F1	Macro-F1
Controlled	TF-IDF	0.8778	0.8889	0.8765
Controlled	Signal	**1.0000**	**1.0000**	**1.0000**
HC3	TF-IDF	0.7117	0.7258	0.7109
HC3	Signal	**0.8117**	**0.8232**	**0.8109**
M4	TF-IDF	**0.8446**	**0.8390**	**0.8444**
M4	Signal	0.7820	0.7810	0.7820

Table 2: In-domain performance of the two classical baselines.

5 Results and Analysis

5.1 Classical Baseline Characterization

Before evaluating neural and fusion models, we first characterize the behavior of the two classical baselines. TF-IDF and Signal features represent two different inductive biases: TF-IDF captures lexical and phrase-level patterns, while Signal features capture global statistical and stylometric properties.

Table 2 summarizes the in-domain behavior of TF-IDF and Signal features across Controlled, HC3, and M4. On simpler in-domain benchmarks such as Controlled and HC3, Signal features perform strongly, suggesting that predictability, burstiness, and style-level regularities are useful when the data distribution is relatively clean or centered on a single generator. However, on the more heterogeneous M4 benchmark, TF-IDF becomes the stronger classical baseline. This indicates that lexical artifacts can remain highly predictive when training and testing occur within the same mixed multi-domain and multi-generator distribution.

This characterization highlights that the two classical baselines capture different but incomplete evidence. TF-IDF provides strong lexical cues, while Signal features provide more global statistical cues. This motivates our fusion design, which combines lexical, statistical, and contextual predictions rather than relying on any single representation.

5.2 Main Classification Results

After characterizing the two classical baselines, we compare them with the fine-tuned RoBERTa neural baseline and the fusion models on the main M4 evaluation settings. Table 3 reports AI-class F1 and Macro F_1 across in-domain, held-out generator, and held-out domain settings. We use Macro F_1 as the primary metric, while AI-class F1 directly measures how well the model identifies AI-generated text.

Table 3 shows that RoBERTa is the strongest

Setting	Method	AI F1	Macro
M4 → M4	TF-IDF	0.839	0.84
M4 → M4	Signal	0.801	0.80
M4 → M4	RoBERTa	0.964	0.96
M4 → M4	Signal-Aware MLP Gate	**0.973**	**0.97**
Held-out ChatGPT	TF-IDF	0.605	0.76
Held-out ChatGPT	Signal	0.596	0.74
Held-out ChatGPT	RoBERTa	0.764	0.85
Held-out ChatGPT	Signal-Aware MLP Gate	**0.791**	**0.87**
Cross-Domain Wiki	TF-IDF	0.645	0.50
Cross-Domain Wiki	Signal	0.716	0.69
Cross-Domain Wiki	RoBERTa	0.945	0.94
Cross-Domain Wiki	Hierarchical Ensemble	**0.947**	**0.94**

Table 3: Main classification results on M4. AI F1 measures performance on the AI-generated class, while Macro F1 averages performance across human and AI classes.

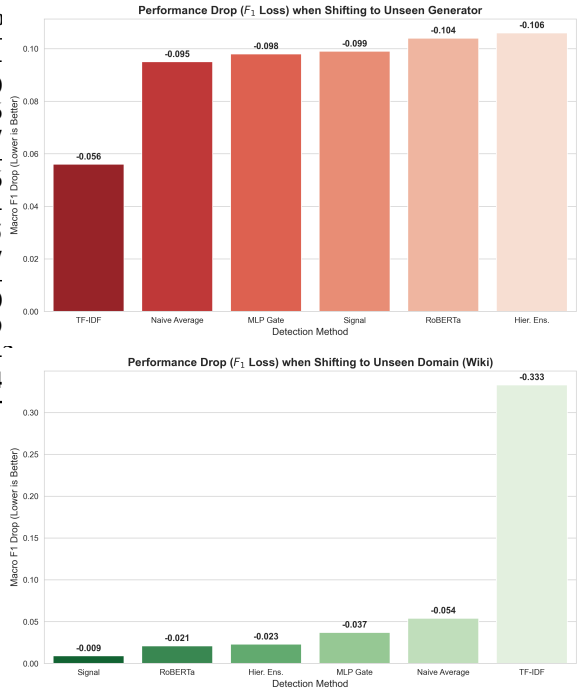


Figure 1: Macro F_1 drop from M4 in-domain evaluation to shifted test settings. Top: held-out ChatGPT generator shift. Bottom: held-out Wiki domain shift. Lower drop indicates better robustness.

standalone detector across all three M4 settings. On M4 in-domain evaluation, RoBERTa reaches 0.963 Macro F_1, substantially outperforming TF-IDF and Signal features. This confirms that contextual transformer representations capture stronger classification evidence than lexical or hand-crafted statistical features alone.

However, fusion further improves performance in the most important settings. In the M4 → M4 setting, the Signal-Aware MLP Gate achieves the best result, improving Macro F_1 from 0.963 for RoBERTa to 0.973. More importantly, in the held-out ChatGPT setting, it improves RoBERTa from 0.859 to 0.875 Macro F_1, and increases AI-class F1 from 0.764 to 0.791. This indicates that the fusion model improves not only overall balanced performance, but also the ability to identify AI-generated text from an unseen generator.

The Cross-Domain Wiki setting shows a different pattern. TF-IDF performs poorly in Macro F_1, suggesting that lexical features are highly sensitive to domain vocabulary. Signal features perform better than TF-IDF but remain much weaker than RoBERTa. The Hierarchical Ensemble achieves the best score in this setting, with 0.945 Macro F_1, slightly improving over RoBERTa. This suggests that different fusion structures may be useful under different types of distribution shift.

5.3 Robustness under Distribution Shift

While the absolute performance in Table 3 shows which models perform best in each setting, it does not fully characterize how stable each detector is when the test distribution changes. We therefore analyze the Macro F_1 drop from the M4 in-domain setting to shifted evaluation settings. Figure 1 visualizes two types of distribution shift: held-out ChatGPT generator shift on the left and held-out Wiki domain shift on the right. Lower drop indicates better robustness.

5.3.1 Generator Shift

The left side of Figure 1 shows that held-out generator evaluation causes a broad performance drop across nearly all methods. Even RoBERTa, the strongest standalone detector in Table 3, drops by about 0.10 Macro F_1. The Hierarchical Ensemble shows a similar drop, while the Signal-Aware MLP Gate and Naive Average also lose close to 0.10. This indicates that generator shift is not only a weakness of shallow lexical models; even strong neural and fusion models remain sensitive to unseen generator behavior.

Interestingly, TF-IDF shows the smallest drop in this setting. However, this should not be interpreted as TF-IDF being the best detector overall, because its absolute held-out ChatGPT performance is still lower than RoBERTa and the Signal-Aware MLP Gate. Instead, the smaller drop suggests that some lexical cues learned from the M4 training generators still transfer to ChatGPT. The stronger absolute

performance of the Signal-Aware MLP Gate suggests that fusion is still beneficial: signal features do not replace RoBERTa, but they provide complementary statistical evidence that helps the fusion model perform better when RoBERTa becomes less reliable under generator shift.

5.3.2 Domain Shift

The right side of Figure 1 reveals a different pattern. Under the held-out Wiki domain shift, TF-IDF suffers the largest drop, decreasing by more than 0.33 Macro F_1. This is much larger than the drop of the other models and shows that lexical n-gram features are highly sensitive to domain vocabulary. When the test domain changes to Wikipedia, the word distributions learned from the training domains no longer transfer well.

In contrast, Signal features show the smallest drop under Wiki domain shift, while RoBERTa and the Hierarchical Ensemble also remain highly stable. This does not mean that Signal is the strongest model in absolute terms; RoBERTa and the Hierarchical Ensemble still achieve much higher Macro F_1 in Table 3. Instead, the small drop suggests that statistical and stylometric cues are less tied to topic-specific vocabulary. Features such as perplexity, burstiness, lexical diversity, and punctuation behavior capture aspects of how text is written rather than only what words appear in the text.

5.3.3 Synthesis

Together, the two plots in Figure 1 show that generator shift and domain shift create different failure modes. Generator shift affects almost all models, including RoBERTa and fusion models, suggesting that detectors can still depend on generator-specific patterns. Domain shift mainly exposes the fragility of lexical memorization, especially for TF-IDF. This supports the motivation for fusion: lexical, statistical, and contextual models capture different evidence and fail in different ways. By combining these views, the fusion models improve shifted performance, even though they do not completely eliminate distribution-shift sensitivity.

5.4 Effect of Fusion Strategies

After analyzing robustness under distribution shift, we examine whether different fusion strategies provide additional benefit beyond the strongest standalone detector. Table 4 reports the Macro F_1 gain of each fusion method over fine-tuned RoBERTa. Positive values indicate that fusion improves over the neural baseline, while negative values indicate that fusion hurts performance.

Method	M4 → M4	Held-out ChatGPT	Wiki
Naive Avg.	+0.002	+0.011	-0.031
MLP Gate	**+0.010**	**+0.016**	-0.006
Hier. Ens.	+0.005	+0.003	**+0.003**

Table 4: Macro F_1 gain of fusion methods over RoBERTa.

The Signal-Aware MLP Gate provides the largest improvement in the M4 → M4 and held-out ChatGPT settings. Its gain under held-out ChatGPT is especially important because this setting directly measures cross-generator robustness. Unlike Naive Average, which assigns fixed equal weights to all base detectors, the MLP Gate receives both base-model probabilities and raw signal features. This allows it to learn interactions between RoBERTa's contextual prediction and stylometric cues such as perplexity, burstiness, lexical diversity, and punctuation behavior. One interpretation is that these signal features act as reliability cues: when RoBERTa is less reliable under an unseen generator, the MLP Gate can use statistical evidence to moderate the final prediction rather than blindly trusting the neural probability.

Naive Average shows the limitation of non-adaptive fusion. It improves slightly over RoBERTa on M4 → M4 and held-out ChatGPT, but it hurts performance on Cross-Domain Wiki. This pattern reflects asymmetric degradation under distribution shift: TF-IDF collapses under Wiki domain shift, while RoBERTa remains relatively stable. Since Naive Average has no mechanism to detect that one base detector has failed, it gives equal weight to a degraded lexical prediction and therefore pulls the final prediction away from the stronger neural model.

The Hierarchical Ensemble performs best on Cross-Domain Wiki, although the gain over RoBERTa is small. This suggests that multi-stage fusion may be helpful when domain shift changes the reliability of the base detectors. Under severe lexical mismatch, a flat fusion model may directly expose the final classifier to noisy lexical evidence. In contrast, the hierarchical structure first combines TF-IDF and Signal predictions through intermediate voters before integrating them with RoBERTa-based predictions. This intermediate layer can be viewed as a feature bottleneck that filters and compresses classical evidence before final fusion.

Overall, these results show that fusion is not automatically beneficial simply because more models are combined. The effectiveness of fusion depends on whether the fusion mechanism can account for different failure modes. Adaptive gating is most useful under generator shift, while hierarchical aggregation provides a small advantage under Wiki domain shift. This supports our broader conclusion that lexical, statistical, and contextual evidence are complementary, but they must be combined in a shift-aware way.

6 Discussion

6.1 Key Findings

Our results suggest three main findings. First, contextual neural representations are the strongest standalone signal for AI-generated text detection. RoBERTa consistently outperforms the lexical and signal-only baselines, showing that semantic and discourse-level representations are important for this task.

Second, hand-crafted signal features are weaker as standalone detectors but remain useful as complementary evidence. They capture statistical and stylometric properties that are less directly tied to specific words, making them valuable in fusion settings.

Third, robustness depends on the type of distribution shift. Generator shift affects even strong neural detectors, while domain shift mainly exposes the weakness of lexical memorization. This suggests that robust AI-text detection requires combining multiple views of text rather than relying on a single representation.

6.2 Error Analysis

We conduct an instance-level error analysis on the held-out ChatGPT setting, comparing RoBERTa with the Signal-Aware MLP Gate. The results reveal that the fusion model improves performance in a counter-intuitive way. Rather than detecting more unseen AI-generated text, the MLP Gate mainly reduces false positives.

RoBERTa produces 350 false negatives, while the MLP Gate produces 352. In addition, the MLP Gate does not recover any of the 350 ChatGPT-generated Wikipedia samples missed by RoBERTa. This suggests that the hardest AI-generated Wikipedia examples remain difficult for both models.

The gain instead comes from false positives. RoBERTa incorrectly classifies 321 human examples as AI-generated, while the MLP Gate reduces this number to 223. Thus, the fusion model corrects 98 human-written examples that RoBERTa falsely flags as AI, without introducing new false positives in this comparison. Many of these corrected examples come from highly structured human writing, especially ArXiv-style academic texts.

This suggests that signal-aware fusion acts primarily as a precision anchor under generator shift. It does not simply catch more unseen AI text; instead, it prevents the neural detector from overflagging formal and formulaic human writing. This is important for practical AI-text detection, where false accusations against human authors can be especially harmful.

6.3 Limitations

Our work has several limitations. First, our main robustness analysis focuses on a subset of M4 and selected held-out settings, especially held-out ChatGPT and Cross-Domain Wiki. A broader evaluation across all generators, domains, and newer LLMs would provide a more complete robustness picture.

Second, RoBERTa inputs are truncated to 512 tokens, which may discard useful evidence from longer documents. Future work could explore chunking or long-context models.

Third, although signal features improve fusion, we do not directly interpret the learned MLP weights or calibrate model uncertainty. Therefore, our explanation of signal features as reliability cues should be understood as an empirical interpretation rather than a fully verified mechanism.

Finally, we do not evaluate adversarially paraphrased or human-edited AI text. Since real-world AI-generated text is often revised after generation, future work should test whether the proposed fusion methods remain robust under paraphrasing and editing.

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